Homework 4

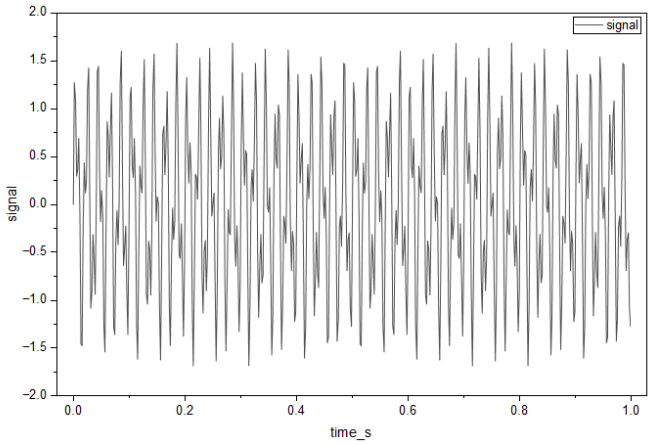
Part 1

1. An FFT is a mathematical operation that analyzes a signal in the frequency domain, showing the characteristic frequencies present in the signal and their amplitudes.
2. - The sampling rate represents the number of samples of a signal acquired every second.
   - The Nyquist frequency is the maximum frequency that can be correctly represented for a given sampling rate, and it is equal to half of the sampling rate.
   - The frequency resolution is the distance between two consecutive frequencies analyzed by the FFT, and it is calculated as the sampling rate divided by the number of samples N.
3. A uniformly spaced time grid is important for FFT analysis because it provides a constant sampling rate, which is necessary for the FFT to work correctly.

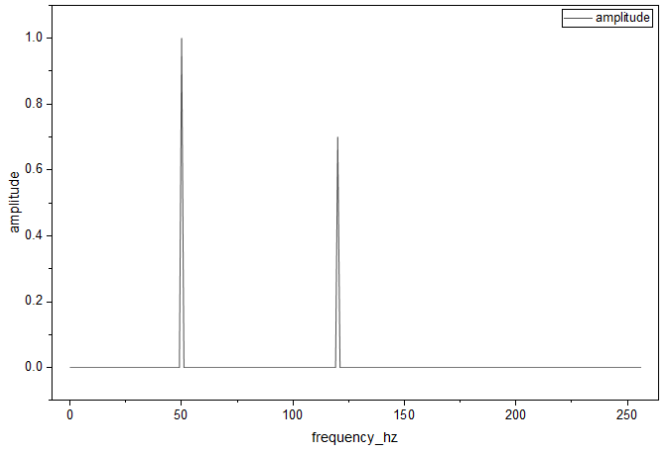
Part 2

3. The program produces eight CSV files:

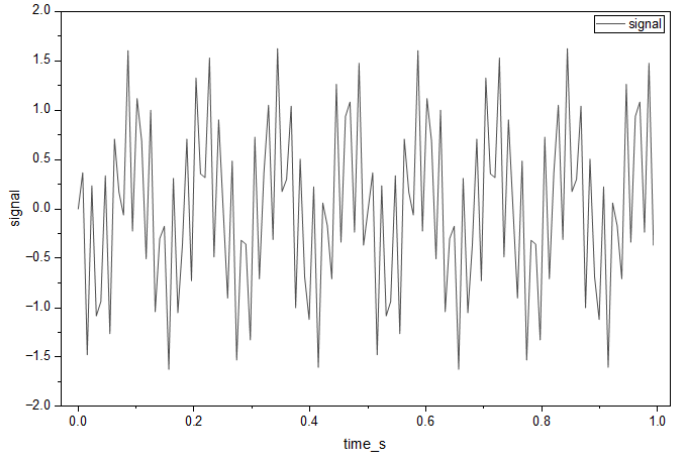
i. good_sampling_signal.csv → It represents the signal in the time domain in the well-sampled case.



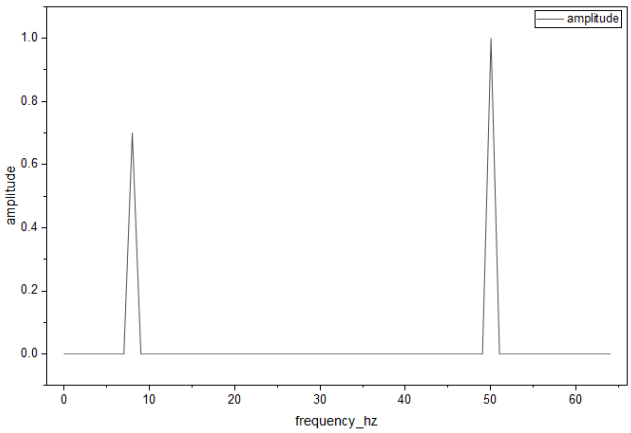
ii. good_sampling_spectrum.csv → It represents the FFT spectrum of the well-sampled signal. The two frequencies reported in the well-sampled case are 50 Hz and 120 Hz.



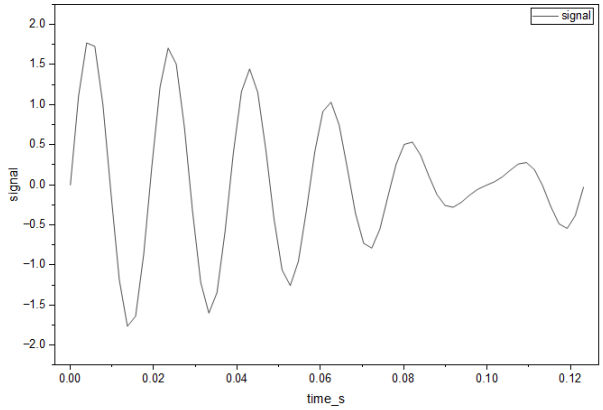
iii. undersampled_signal.csv → It represents the same signal in the time domain, but with a lower sampling rate.



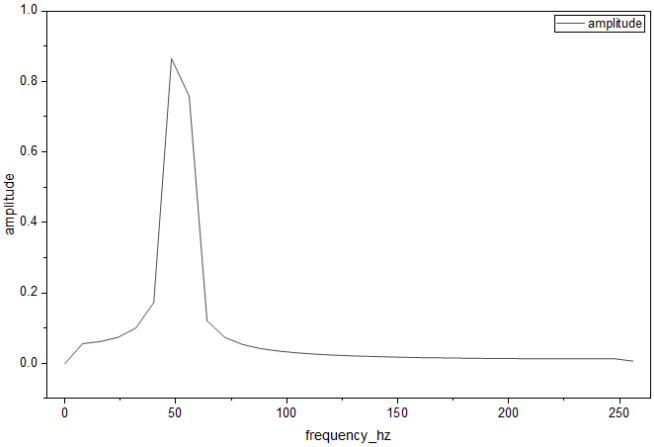
iv. undersampled_spectrum.csv → It represents the FFT spectrum of the undersampled signal.



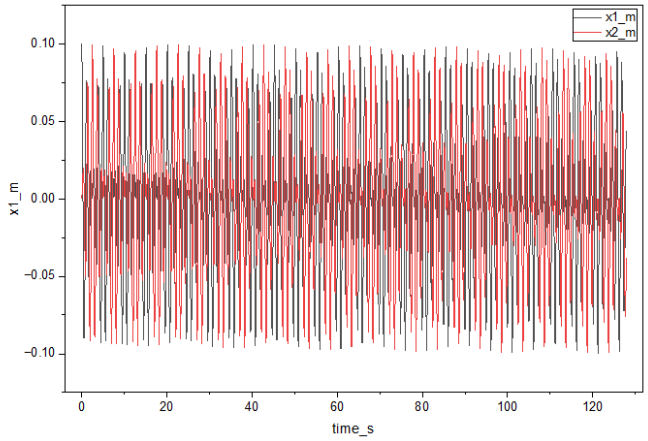
v. short_record_signal.csv → It represents a signal observed for a short time interval of 0.125 s.



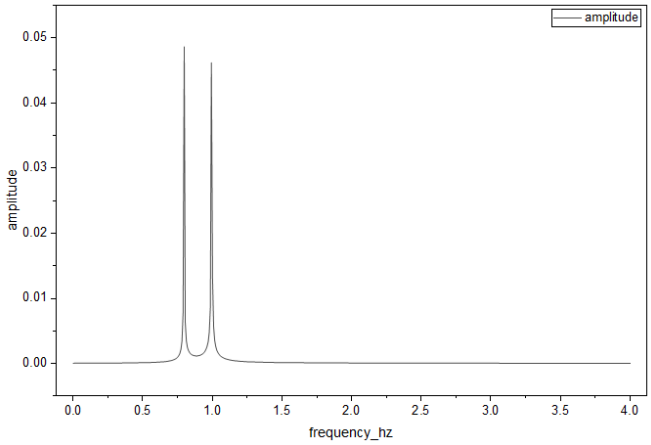
vi. short_record_spectrum.csv → It represents the FFT spectrum of the short-record signal.



vii. coupled_oscillators_time.csv → It represents two coupled oscillators, x_1 and x_2, as functions of time.



viii. coupled_oscillators_spectrum.csv → It represents the FFT spectrum of the coupled oscillators.



Part 3

3. The 120 Hz component cannot be reconstructed faithfully because it is above the Nyquist frequency of 64 Hz.
4. Physically, if the sampling rate is too low, the samples are too far apart to follow the fast oscillation correctly. The sampled points can then look like a slower oscillation. Mathematically, frequencies separated by multiples of the sampling frequency can produce the same sampled sequence. In this case, 120 Hz sampled at 128 Hz appears as 128 - 120 = 8 Hz.

Part 4

2. The total record length is T = 0.125 s
3. The frequency spacing of the FFT bins is 8 Hz.
4. The two frequencies are difficult to separate because they are only 5 Hz apart, while the frequency resolution is 8 Hz. Therefore, the FFT resolution is not sufficient to clearly distinguish the two frequencies.
5. I would increase the total acquisition time. Since $\Delta f = 1/T$, increasing the acquisition time decreases the frequency spacing and improves the frequency resolution. This makes it easier to separate the 50 Hz and 55 Hz components.

Part 5

1. The equations are:

$$\ddot{x}_1 = -\frac{k}{m}x_1 - \frac{k_c}{m}(x_1 - x_2)$$
$$\ddot{x}_2 = -\frac{k}{m}x_2 - \frac{k_c}{m}(x_2 - x_1)$$

2. The equations describe two masses connected to springs and coupled by another spring.

3. The system has two normal modes: an in-phase mode, where the masses move in the same direction, and an out-of-phase mode, where they move in opposite directions.
4. The FFT of x_1 shows two important frequencies because the motion contains both normal modes.
5. Comparing the FFT peaks with the theoretical frequencies is useful to verify that the numerical simulation agrees with the physical theory.

Part 6

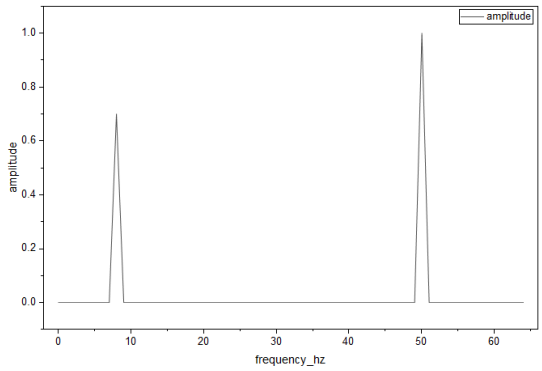
1. The undersampled spectrum makes aliasing easiest to see because the original 120 Hz component appears as a false peak at 8 Hz.
2. The short-record spectrum makes the limited frequency resolution easiest to see because the 50 Hz and 55 Hz components cannot be clearly separated.
3. Since:

	In phase mode	**Out of phase mode**
expected	0.79577 Hz	0.99392 Hz
measured	0.79688 Hz	0.99219 Hz

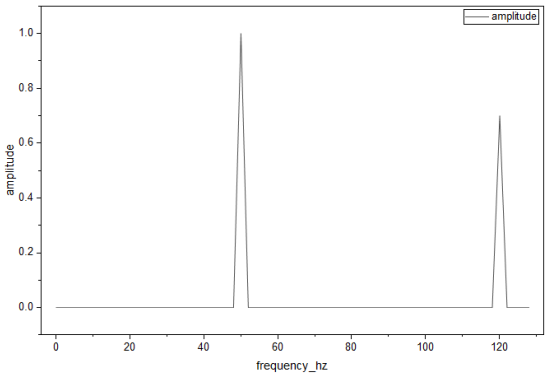
the numerical FFT peaks line up very well with the theoretical reference frequencies
4. The small differences are mainly due to the finite frequency resolution of the FFT.

Part 7

Changing the sampling rate from 128 to 256.
Before:



After:



The FFT output agrees with my prediction. After increasing the sampling rate from 128 Hz to 256 Hz, the peaks appear at 50 Hz and 120 Hz. The alias peak at 8 Hz disappears because the Nyquist frequency is now 128 Hz, which is higher than both signal frequencies.

Part 8

Thanks to this project, I have learnt that analysing a signal is not a trivial task, but requires a thorough understanding of how the modelling system works in order to understand if the analysis is numerically stable. In particular, I have learnt how the sampling rate and acquisition time can influence the results of an FFT, especially in terms of aliasing and frequency resolution. I would like to expand on this project by applying this computational model to various physical systems and studying their behaviour through FFT analysis.